QUALITY OF SERVICE (QOS) SPECIFICATION

SystemVerilog, UVM - training, methodology consulting, auditing & debug

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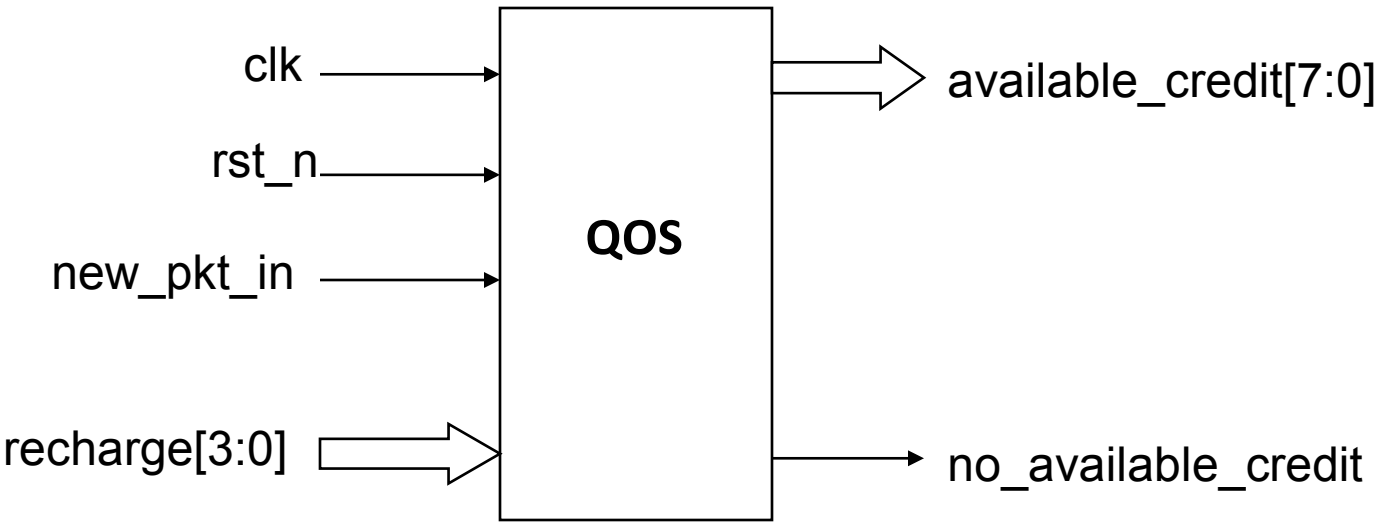
QOS BLOCK:



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Signal Description:

SIGNAL	DIRECTION	WIDTH	DESCRIPTION
clk	Input	1	Clock 50 MHz.
rst_n	Input	1	Active low, asynchronous reset
new_pkt_in	Input	1 bit	Indicates the start of packet.
recharge	Input	4 bit	Recharge signal.
available_credit	output	8 bit	Credit output.
no_available_credit	output	1 bit	Indicates whether the available credit is zero or not.

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EXPLANATION :

- Whenever "new_pkt_in" is asserted the available credit is decremented by 1.
- Whenever "available_credit" becomes zero, "no_available_credit" is asserted.
- After every five clock cycles, the "recharge" value is added to "available_credit".
- The maximum & minimum values of "available_credit" are 255 & 0 respectively.

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EXPLANATION (Contd..) :

- When "available_credit" reaches its maximum value, it should not go beyond even if "recharge" is asserted.
- When "available_credit" reaches its minimum value, it should not go below zero even if "new_pkt_in" is asseted.

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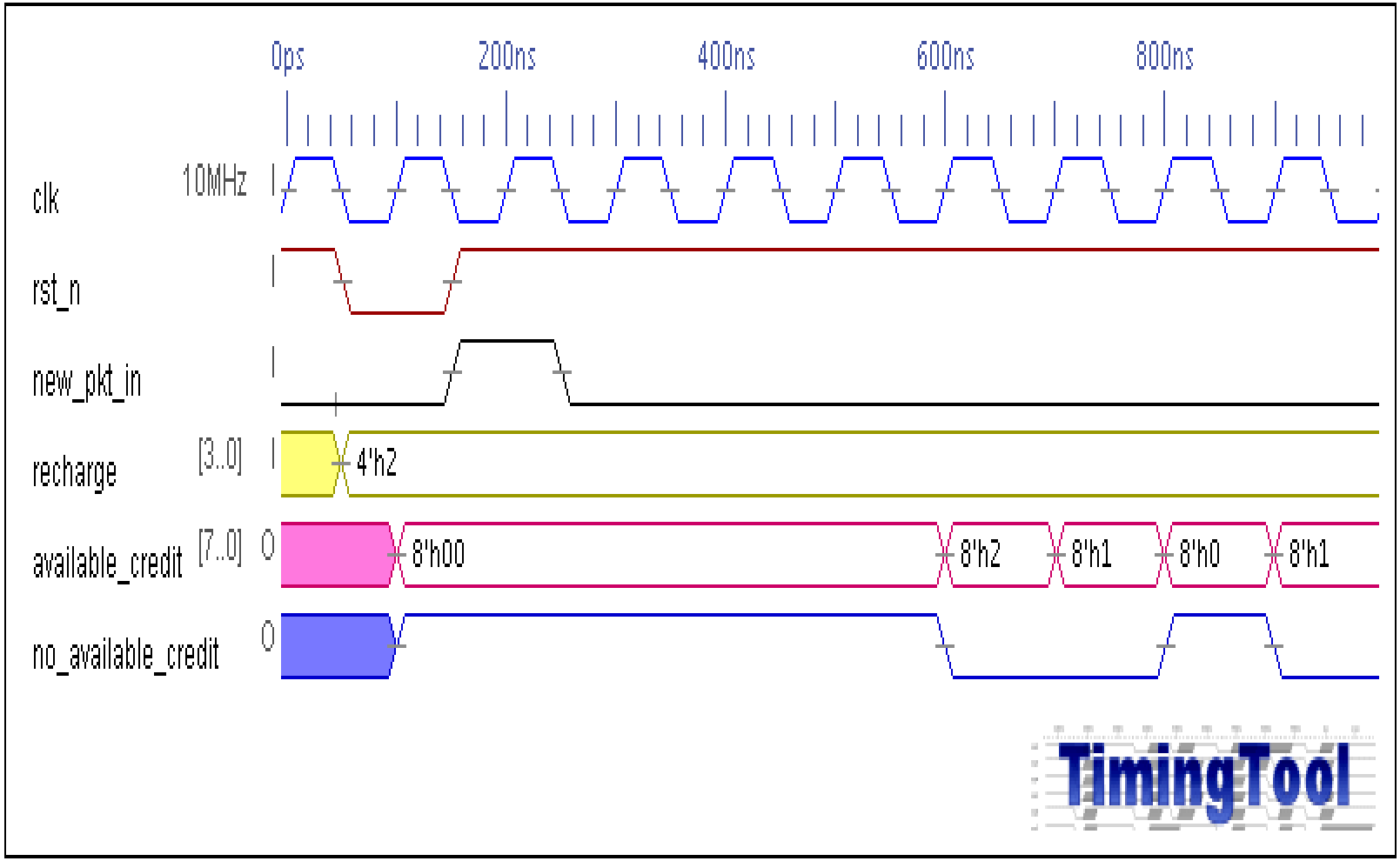
Timing_Diagram



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